

Report

Of

Development of Cybersecurity Utility Tools: Password Strength Evaluator, Vulnerability Scanner, Phishing Email Detection Model, and Secure Login System

Submitted

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By

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CANDIDATE'S DECLARATION

I, ABHINAV ANAND, do hereby declare that the internship report titled “Development of Cybersecurity Utility Tools: Password Strength Evaluator, Vulnerability Scanner, Phishing Email Detection Model, and Secure Login System” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work, particularly in the areas of password security, network vulnerability assessment, machine learning based threat detection, and secure authentication design that this internship covered.

I express my gratitude to my parents for their blessings.

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Instruction for the report format

- Cover Page, Copy of the Certificate issued by industry/organisation will have no Page Number.
- Page no. of Sr. No. 1 to 6 along with a list of tables and list of figures will bear the Roman Numbering System (e.g. i, ii, iii).
- Bibliography/References should be in proper format.

3. Internship Completion Certificate



4. Project Description

4.1 Introduction

Cybersecurity has become one of the most critical concerns in the modern digital landscape, as individuals and organisations face a rapidly growing range of threats — from weak authentication practices to network intrusions, phishing campaigns, and unauthorised account access. During this internship, four complementary cybersecurity utility tools were studied, designed and implemented: a Password Strength Evaluator, a Vulnerability Scanner, a Phishing Email Detection Model, and a Secure Login System. Together these four modules address different layers of information security — the strength of individual user credentials, the security posture of a network or web application, the human-targeted threat of phishing, and the technical safeguarding of the login process itself. This report documents the objectives, design, implementation and outcomes of each of the four sub-projects, along with the concepts of applied cryptography, network scanning, machine-learning based text classification, and secure session management that were studied over the course of the internship.

4.2 Organization Profile

This internship was undertaken at [Name of Organisation/Company], [one or two lines describing the organisation — its area of work, e.g. IT services, cybersecurity training, or software development, and its location]. The organisation provided guidance and a working environment for exploring practical cybersecurity concepts, including secure coding practices, common web application vulnerabilities, and the use of open-source security and machine-learning tooling. The internship was carried out under the mentorship of [Mentor/Supervisor Name, Designation] over a period of [duration, e.g. 6 weeks], during which the four projects described in this report were completed.

4.3 Problem Statement

1. Password Strength Evaluator — Users frequently choose weak, short, or reused passwords, which remain one of the leading causes of account compromise. There is a need for a tool that can evaluate password strength in real time and guide users toward safer choices.

2. Vulnerability Scanner — Web applications and networks often run outdated software or expose open ports and weak configurations without the administrator's knowledge, leaving them susceptible to exploitation. A lightweight scanner is needed to surface these issues early.
3. Phishing Email Detection Model — Phishing remains one of the most common attack vectors, tricking users through deceptive emails that mimic legitimate senders. Manual identification is unreliable at scale, motivating an automated, ML-based classification approach.
4. Secure Login System — Many login systems still store passwords insecurely or fail to validate input properly, exposing them to credential theft and injection attacks. A properly hashed, validated, and session-managed login system is required.

4.4 Project Objectives

Password Strength Evaluator:

- Check password length, character complexity (uppercase, lowercase, digits, symbols), and uniqueness.
- Suggest stronger password alternatives when a weak password is entered.
- Optionally integrate with a database of previously used passwords to prevent reuse.

Vulnerability Scanner:

- Scan a target host or network range for open ports and weak/default configurations.
- Identify outdated software versions running on discovered services.
- Generate a simple, readable vulnerability report summarising findings.

Phishing Email Detection Model:

- Train a machine-learning model on a labelled dataset of phishing and legitimate emails.
- Extract and analyse email features such as embedded URLs, sender patterns, and suspicious keywords.

- Classify unseen emails as “Phishing” or “Safe” and report accuracy and confusion matrix.

Secure Login System:

- Implement user registration and login using salted password hashing (bcrypt or Argon2).
- Apply input validation and parameterised queries to prevent SQL injection.
- Provide session management with a secure logout feature, with optional Two-Factor Authentication (2FA).

4.5 Scope of the Project

The scope of this internship was limited to building functional, educational-grade prototypes rather than production-ready security products. The Password Strength Evaluator and Secure Login System were built as standalone/web-based demonstrations; the Vulnerability Scanner was scoped to scan systems and networks that the student had explicit permission to test (e.g. local/lab environments), consistent with responsible and ethical security practice; and the Phishing Email Detection Model was scoped to publicly available, pre-labelled email datasets rather than live email traffic. The combined scope gives a rounded, hands-on introduction to authentication security, network reconnaissance, applied machine learning for security, and secure web development.

4.6 Technologies and Tools Used

Project	Technologies / Tools
Password Strength Evaluator	Python, regular expressions (re module), zxcvbn / custom scoring logic, SQLite (for reuse-check database)
Vulnerability Scanner	Python, nmap / python-nmap, socket programming, banner grabbing for version detection
Phishing Email Detection Model	Python, Scikit-learn, Pandas, NumPy, NLTK (text preprocessing), TF-IDF vectorisation, Matplotlib/Seaborn for confusion matrix
Secure Login System	Python (Flask/Django) or Node.js, bcrypt / Argon2, HTML-CSS-JavaScript, SQLite/MySQL, session cookies, optional TOTP-based 2FA

4.7 System Architecture

Password Strength Evaluator:

User Input (password) → Rule-based/Entropy Scoring Engine → Strength Score & Feedback → (optional) Lookup against stored password-history database → Suggested stronger alternative displayed to the user.

Vulnerability Scanner:

Target Input (IP/hostname/range) → Port Scanning Module → Service/Version Fingerprinting → Known-vulnerability Matching → Report Generator (text/HTML output).

Phishing Email Detection Model:

Raw Email Dataset → Preprocessing (cleaning, tokenisation) → Feature Extraction (URL patterns, keyword frequency, TF-IDF) → Classifier (e.g. Logistic Regression / Random Forest / Naive Bayes) → Prediction (Phishing/Safe) → Evaluation (Accuracy, Confusion Matrix).

Secure Login System:

Client (Registration/Login Form) → Input Validation Layer → Password Hashing (bcrypt/Argon2) → Database (hashed credentials only) → Session Manager → Optional 2FA Verification → Authenticated Dashboard / Logout.

4.8 Methodology

1. Requirement study — reviewed common password policies, OWASP guidelines, and existing phishing/vulnerability datasets to understand realistic threat patterns.
2. Design — defined the architecture and data flow for each of the four modules, as summarised in Section 4.7.
3. Implementation — built the Password Strength Evaluator and Secure Login System as working code; implemented the port-scanning logic for the Vulnerability Scanner; trained and evaluated the Phishing Email Detection Model on a labelled dataset.
4. Testing — tested the password evaluator against a range of weak and strong sample passwords; ran the vulnerability scanner against a local/lab test environment; validated the ML model using a held-out test split and confusion

matrix; tested the login system for hashing correctness, input validation, and session expiry.

5. Documentation — recorded findings, screenshots, accuracy metrics and vulnerability reports for inclusion in this internship report.

4.9 Expected Outcomes

Password Strength Evaluator:

A working understanding of password security and basic cryptographic concepts, demonstrated through a tool that reliably scores password strength and nudges users toward stronger, non-reused passwords.

Vulnerability Scanner:

Practical insight into penetration testing and vulnerability assessment, evidenced by a script that can identify open ports, weak configurations, and outdated software versions, and summarise them in a simple report.

Phishing Email Detection Model:

A model that classifies emails as “Phishing” or “Safe” with high accuracy on the test dataset, based on textual content and URL features, together with an understanding of feature engineering for security-focused ML.

Secure Login System:

A login system with hashed passwords, input validation, session management, and optional 2FA that significantly reduces the attack surface for unauthorised access compared to a naive login implementation.

4.10 Certificates of Completion and Communication Proof

[Attach the internship completion certificate and any relevant email/communication proof from the organisation here as scanned images or embedded objects.]

5. Bibliography/References

1. OWASP Foundation, "OWASP Testing Guide" and "OWASP Top Ten," available at: <https://owasp.org>
2. NIST Special Publication 800-63B, "Digital Identity Guidelines — Authentication and Lifecycle Management," National Institute of Standards and Technology.
3. Scikit-learn Developers, "Scikit-learn: Machine Learning in Python," documentation available at: <https://scikit-learn.org/stable/documentation.html>
4. Python Software Foundation, "Python 3 Documentation," available at: <https://docs.python.org/3/>
5. bcrypt / Argon2 project documentation, for password hashing algorithm reference.
6. Nmap Project, "Nmap Network Scanning — Official Documentation," available at: <https://nmap.org/book/>

Note: the above references are commonly cited, publicly available sources for the concepts used in this project. Please verify each link/citation before final submission, as exact editions or URLs may need updating.